

Theoretical and Empirical Analysis of Option Greeks and Implied Volatility

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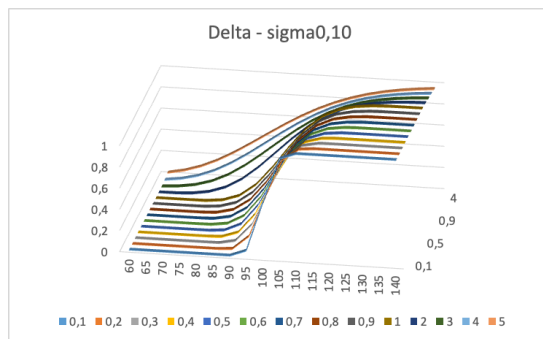
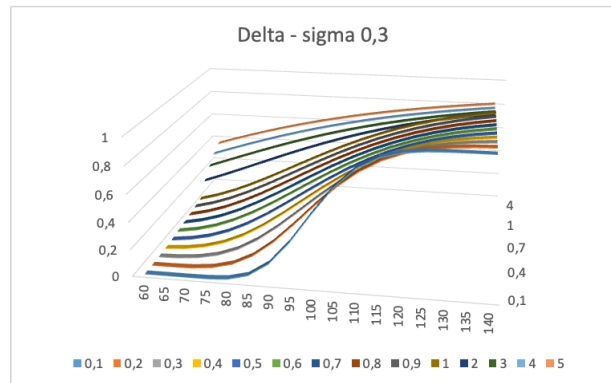
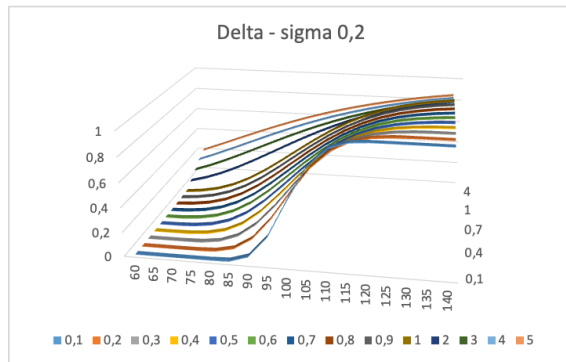
Executive Summary

This report provides a comprehensive analysis of option pricing dynamics using the Black-Scholes-Merton (BSM) framework. The study is divided into three sections: a theoretical sensitivity analysis of Greeks under volatility shocks, an empirical investigation of Stellantis N.V. market data, and a cross-validation between a custom-built VBA model and professional LSEG (Refinitiv) benchmarks. Key findings highlight the impact of high dividend yields on implied volatility and the necessity of unit-scaling when comparing theoretical models with market terminals.

Theoretical Greeks and Volatility Shocks

The first part of the study examines how changes in volatility (σ) affect the sensitivity of option prices. Using a base volatility of 20% and shocks of $\pm 50\%$ (10% and 30%), we mapped the Greeks across different strikes and maturities. To compute the greeks we used a VBA script from [VinegarHill-FinanceLabs](#).

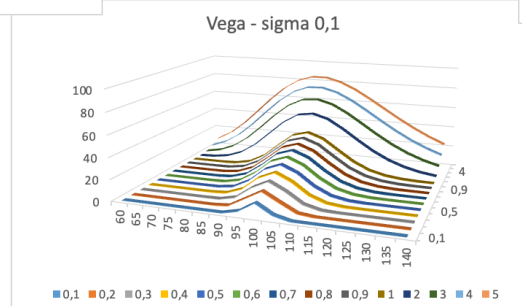
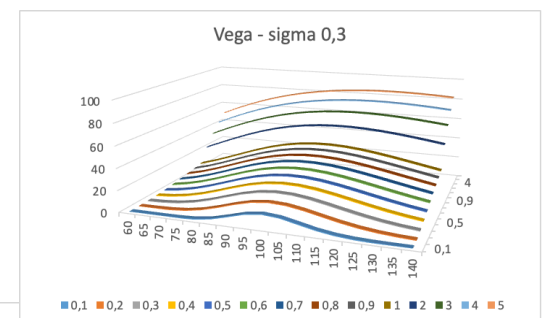
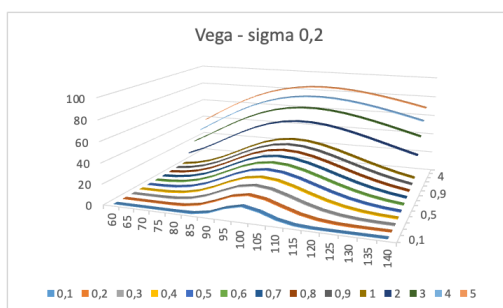




A central finding of this theoretical analysis is the **General Smoothing Effect** induced by higher volatility. Higher volatility induces a "smoothing effect" across all Greeks by spreading the terminal price distribution; as σ increases, the Gamma peak flattens and widens, making the Delta transition less abrupt and the overall surfaces more stable.

Simultaneously, Vega remains highest for at-the-money options and increases with time to maturity, representing the option's fundamental sensitivity to shifts in market uncertainty.

As maturity increases, the Greeks undergo a "diffusion" effect: the Delta slope becomes gentler and the Gamma peak flattens significantly, reflecting a wider distribution of potential price outcomes. Conversely, Vega and Rho rise as longer durations amplify the option's sensitivity to volatility and interest rate shifts.

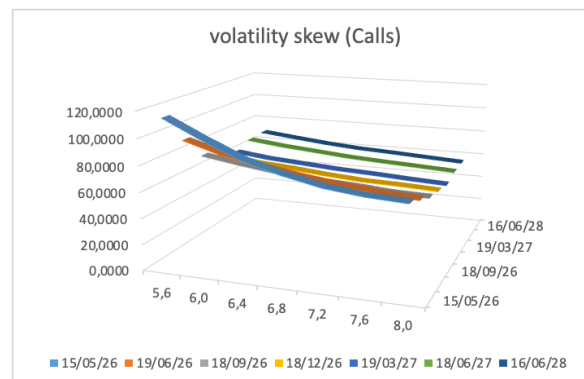


Empirical Analysis – The Volatility Surface

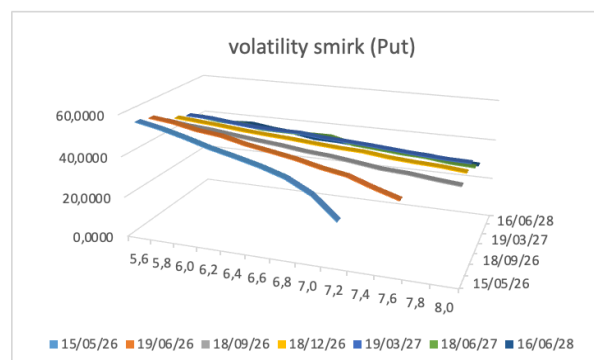
Stellantis N.V. was selected as the primary underlying for the empirical analysis. As a leading global automaker (resulting from the merger of FCA and PSA), the stock is characterized by a cyclical nature and a significant dividend policy. At the time of analysis, the stock traded at approximately **€6.80** with an expected annual dividend of **€0.68**, resulting in a high **Dividend Yield (\$q\$) of 10%**. This parameter is crucial for adjusting the standard Black-Scholes model to ensure accurate pricing and Greek calculation. (I realised too late that it yields dividends).

Using market data extracted via LSEG Workspace, we reconstructed the Implied Volatility (IV) surface for Stellantis.

The Call surface displays a traditional **convex** decay. The implied volatility decreases as the strike rises, but the rate of decrease slows down, creating a stable, "bowed" shape toward the origin (**Skew**).



Interestingly, the Put surface exhibits a **concave downward profile**. Instead of a gradual decline, the implied volatility "accelerates" its descent as the strike moves from deep-out-of-the-money towards the spot price. This **concavity** is a direct empirical signature of the "Dividend Cliff": the market aggressively reprices the volatility risk as the options approach the expected ex-dividend price drop, creating a steep, dome-like trajectory rather than a smooth smile.



As time to maturity increases, the volatility smile becomes progressively flatter because short-term market shocks and discrete event risks average out over a longer horizon.

A notable finding was the elevated IV levels (approx. 60%) for mid-term maturities. This is primarily an artifact of the discrete dividend stake.

The relatively elevated implied volatilities are plausible because:

1. The automotive sector is cyclical and sensitive to macroeconomic uncertainty.
2. Stellantis is exposed to EV transition risk, regulation, and global demand fluctuations.
3. **Dividend payments matter:** when dividends are expected, forward prices are lower than spot prices, so using a dividend-adjusted Black–Scholes model is necessary.

Smoothness of Greeks - Empirical Vs VBA

The custom VBA engine was tested against Refinitiv benchmarks for the September 2026 maturity. Using $q=0.1\%$ the results were approximately the same.

A direct comparison initially showed large discrepancies in **Vega, Rho, and Theta**. However, this was identified as a difference in **units of measurement**:

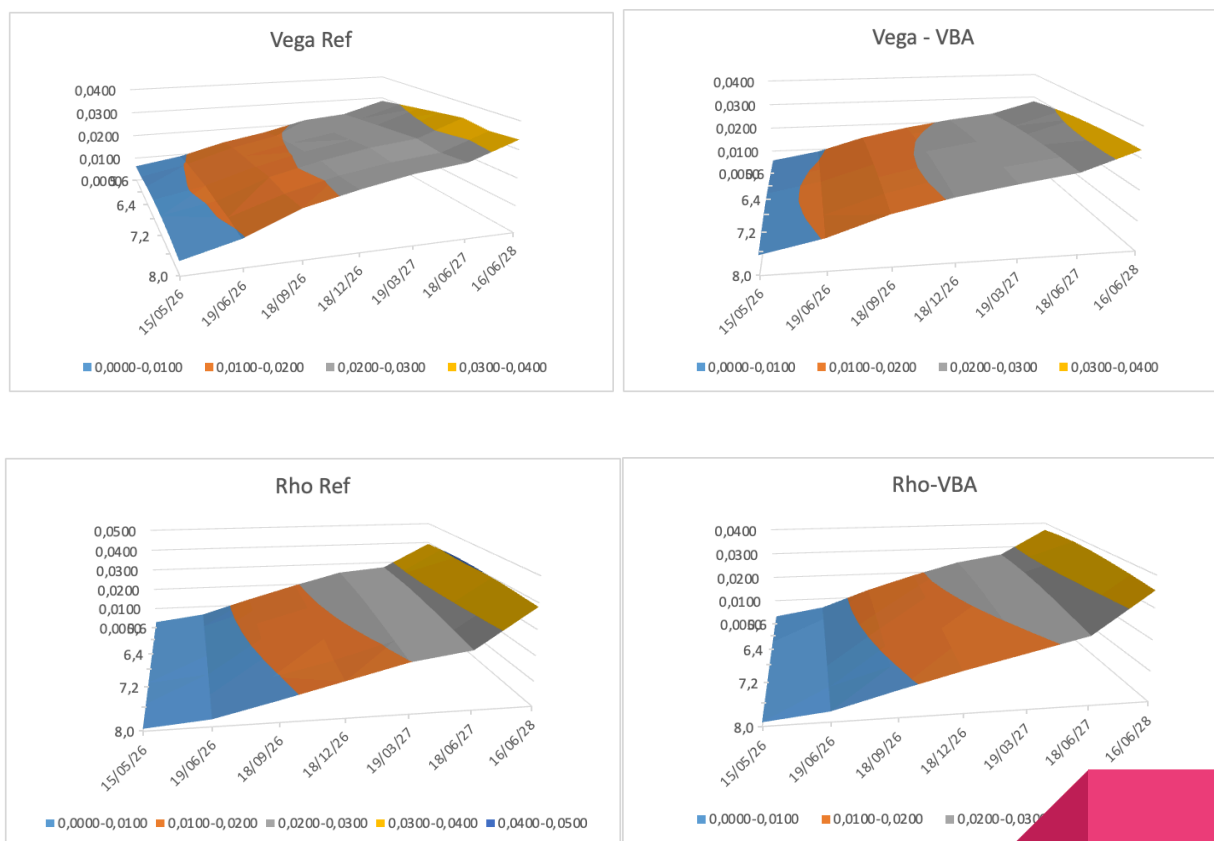
	DeltaC	gamma	Vega	ThetaC	RhoC
VBA	0,53690527	0,13667523	1,7069424	-0,9095232	1,16276368
Refinitiv	0,5570	0,1606	0,0167	-0,0035	0,0115

- **Vega & Rho:** VBA calculates sensitivity for a 100% (1.0) move in σ or r . Refinitiv reports sensitivity for a 1% (0.01) move. Dividing VBA results by 100 aligned the values perfectly.

- **Theta:** VBA computes an annual decay. Refinitiv reports a **daily decay**. Dividing the VBA output by **252** matched the market data.

A key qualitative difference observed is the "smoothness" of the Greek surfaces.

- **VBA Theoretical Smoothness:** The Greeks generated via the VBA script exhibit a high degree of **analytical smoothness**. This is expected as the Black-Scholes-Merton model relies on continuous probability distributions. The smoothness would be more evident if more data were to be used.
- **Refinitiv Empirical "Jaggedness":** In contrast, the Greeks reported by Refinitiv often exhibit minor "noise" or irregularities. This is because they are derived directly from **actual market prices**, which are subject to:
 1. **Bid-Ask Spreads:** The gap between buying and selling prices creates small "kinks" in the implied data.
 2. **Liquidity Constraints:** Options with low trading volume may have prices that don't perfectly align with the theoretical curve.
 3. **Discrete Strike Spacing:** Market data exists only at specific intervals, forcing the terminal to use local approximations that can result in a less "liquid" look compared to the VBA analytical ideal.



The characteristic "textbook" shapes of the Greeks are less apparent here due to the exceptionally high implied volatility levels. All considered, the graphs are similar, which indicates consistency in the work.

CONCLUSIONS

The results obtained are coherent both theoretically and empirically. In Part 1, the Greeks react to volatility exactly as expected under Black–Scholes. In Part 2, Stellantis option data display the classic equity skew/smirk pattern, with elevated implied volatilities justified by market risk and dividend effects. In Part 3, the VBA model closely replicates Refinitiv Greeks, with only limited deviations and slightly smoother outputs. Overall, the project demonstrates a solid implementation of option pricing theory combined with realistic market-data interpretation.